

THEME BASED INTERDISCIPLINARY PROJECT
REPORT ON
Material Property Prediction Using Machine Learning &
Deep Learning Models

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**VASAVI COLLEGE OF ENGINEERING (AUTONOMOUS)
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CERTIFICATE

This is to certify that the Theme based project report entitled “**Material Property Prediction Using Machine Learning & Deep Learning Models**” is a bonafide record of a Theme based project work carried out by

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in the Department of Mechanical Engineering, Vasavi College of Engineering (Autonomous) (Affiliated to Osmania University), Hyderabad during the academic year 2025-26.

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VI ABSTRACT

This project focuses on predicting the mechanical properties of TIG-welded mild steel joints using machine learning and deep learning techniques. The study was carried out based on the ISRO problem statement, “Material Property Prediction Using Machine Learning and Deep Learning Models.” Due to financial limitations, mild steel was selected as a substitute for aerospace-grade materials while maintaining the same experimental and analytical approach.

Mild steel plates of dimensions 30 mm × 150 mm and thickness 3 mm were cut and welded using the TIG welding process. The welding parameters considered in this study were welding current, arc voltage, and shielding gas flow rate. Eleven specimens were prepared using different combinations of these parameters to generate a suitable dataset for analysis.

Each welded specimen was tested using a Universal Testing Machine (UTM). From the tensile tests, load–displacement and stress–strain curves were obtained, and important mechanical properties such as yield strength, ultimate tensile strength, and elongation were determined. These experimental results were used as the input dataset for developing machine learning models.

The developed machine learning and deep learning models are used to predict the mechanical properties of welded joints based on welding parameters. This approach reduces the need for repeated experimental testing and helps optimize welding conditions efficiently. The methodology developed in this project provides a low-cost proof of concept that can be extended to advanced materials used in aerospace and other critical engineering applications.

VII Introduction

TIG welding is widely used in industries because it produces high-quality welds with good control over heat input. The mechanical properties of a welded joint depend mainly on welding parameters such as current, voltage, and shielding gas flow rate. These parameters influence the strength, ductility, and overall performance of the welded material.

In this project, mild steel plates of dimensions $30\text{ mm} \times 150\text{ mm}$ and thickness 3 mm were used due to financial limitations. The plates were joined using TIG welding by varying the welding current, voltage, and gas flow rate. A total of eleven welded specimens were prepared under different parameter combinations to create an experimental dataset.

The welded specimens were tested using a Universal Testing Machine (UTM). From the tensile tests, load–displacement and stress–strain curves were obtained, and important mechanical properties such as yield strength, ultimate tensile strength, and elongation were calculated.

The experimental data were then used to develop machine learning and deep learning models to predict the mechanical properties of welded joints. This approach reduces the need for repeated destructive testing and helps in selecting optimal welding parameters efficiently. The work serves as a cost-effective implementation of the ISRO problem statement on material property prediction using artificial intelligence.

VIII LITERATURE REVIEW

1. EXPERIMENTAL INVESTIGATION AND OPTIMIZATION OF TIG WELDING PARAMETERS ON ALUMINUM 6061 ALLOY USING FIREFLY ALGORITHM - **Rishi Kumar** ¹ **Ramesh N Mevada** ², **Santosh Rathore**³, **Nitin Agarwal**⁴, **Vinod Rajput**⁵. **AjayPal Singh Barad** ⁶

Abstract: To improve Welding quality of aluminum (Al) plate, the TIG Welding system has been prepared, by which Welding current, shielding gas flow rate and Current polarity can be controlled during Welding process. In the present work, an attempt has been made to study the effect of Welding current, current polarity, and shielding gas flow rate on the tensile strength of the weld joint. Based on the number of parameters and their levels, the Response Surface Methodology technique has been selected as the Design of Experiment. For understanding the influence of input parameters on Ultimate tensile strength of weldment, ANOVA analysis has been carried out. Also to describe and optimize TIG Welding using a new metaheuristic Nature - inspired algorithm which is called as Firefly algorithm which was developed by Dr. Xin-She Yang at Cambridge University in 2007. A general formulation of firefly algorithm is presented together with analytical, mathematical modeling to optimize the TIG Welding process by a single equivalent objective function.

2. DEEP LEARNING BASED REAL-TIME PREDICTION OF DEPTH OF PENETRATION DURING ACTIVATED TUNGSTEN INERT GAS WELDING OF 10 MM THICK 316LN STAINLESS STEEL –

N. Chandrasekhar, M. Vasudevan & C. R. Das

Abstract: A deep learning-based approach has been developed for real-time weld pool semantic segmentation and predicting the depth of penetration (DOP) during activated tungsten inert gas (A-TIG) welding of 10 mm thick 316LN stainless steel plate. A custom U-Net framework, with VGG19 as the encoder, was used for semantic segmentation of weld pool images in real-time. Model development involved three main steps: semantic segmentation of the weld pool using a convolutional neural network (CNN), surface profile measurement and DOP prediction using a back-propagation neural network (BPNN). The CNN model exhibited excellent accuracy on the segmented images. Key features were extracted from the segmented images and along with the welding current were used as inputs to the BPNN while the measured DOP values were used as output to train the BPNN. All the three steps were combined to predict the depth of penetration in real-time. The model's execution time was found as 110 ms. Replacing the encoder with Efficient Net-BO reduced the execution time to 70 Ms. A Python-based graphical user interface (GUI) was developed to monitor the depth of penetration as a function of weld distance in real-time. Excellent agreement was achieved between the predicted and measured values.

1X. PROBLEM STATEMENT, OBJECTIVES, AND EXPECTED OUTCOMES

Problem Definition: -

The mechanical properties of TIG-welded joints vary significantly with welding parameters such as current, voltage, and shielding gas flow rate. Determining the optimum combination of these parameters through repeated experimental testing is time-consuming and costly.

The objective of this project is to develop machine learning and deep learning models that can predict tensile properties such as yield strength, ultimate tensile strength, and elongation based on welding parameters.

Objectives of the Project

- To study the effect of TIG welding parameters such as welding current, arc voltage, and shielding gas flow rate on the mechanical properties of mild steel joints.
- To prepare mild steel specimens of dimensions 30 mm × 150 mm × 3 mm and weld them using different combinations of process parameters.
- To conduct tensile tests on the welded specimens using a Universal Testing Machine (UTM).
- To obtain Load–Displacement and Stress–Strain curves from the tensile test results.
- To extract important mechanical properties such as maximum load, tensile strength, and percentage elongation from the experimental data.

- To create a structured dataset containing welding parameters as inputs and mechanical properties as outputs.
- To preprocess the experimental data for machine learning.
- To develop machine learning models such as Linear Regression, Random Forest, and Support Vector Regression.

Expected Outcomes

- Successful fabrication and tensile testing of TIG-welded mild steel specimens under different welding conditions.
- Generation of Load–Displacement and Stress–Strain curves for all test specimens.
- Extraction of mechanical properties such as maximum force, tensile strength, and elongation.
- Development of a reliable experimental dataset for machine learning and deep learning.
- Training and validation of predictive models capable of estimating mechanical properties from welding parameters.
- Identification of the best-performing machine learning or deep learning model.
- Reduction in time, cost, and effort required for repeated destructive testing.
- Determination of the influence of welding current, voltage, and gas flow rate on tensile properties.
- Creation of a predictive framework that can be extended to aerospace materials and other engineering applications.

X EXPERIMENTAL SETUP

X.1 Preparation of Mild Steel Specimens: -



Fig 1 :-Cutting of Specimen



Fig 2:-Grinding of the Specimen



Fig 3: - Cut Mild Steel Specimens

X.2 TIG Welding



Fig 4:- Weld Set Up



Fig 6: - Welded Specimens



Fig 5 Welding Of the Specimens

X.3 Tensile Testing: -



Fig 7 Testing Set Up



Fig 8 Testing of the Specimen

X.4 Extraction of Tensile Test Data

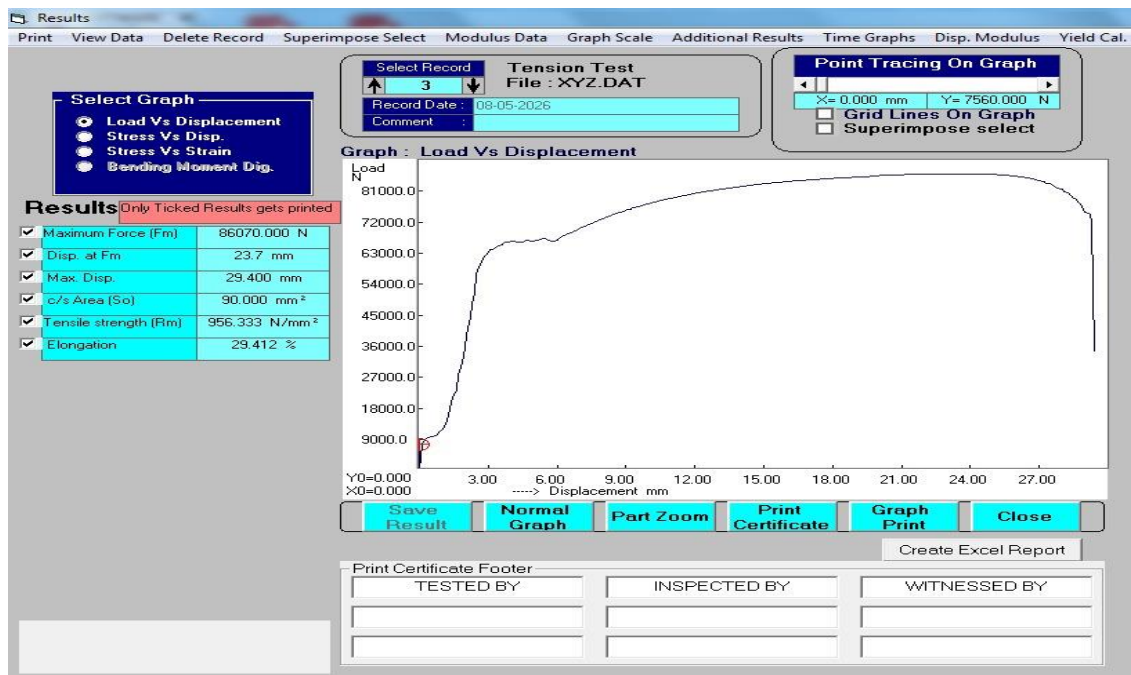


Fig 9: - Load Vs Displacement Graph

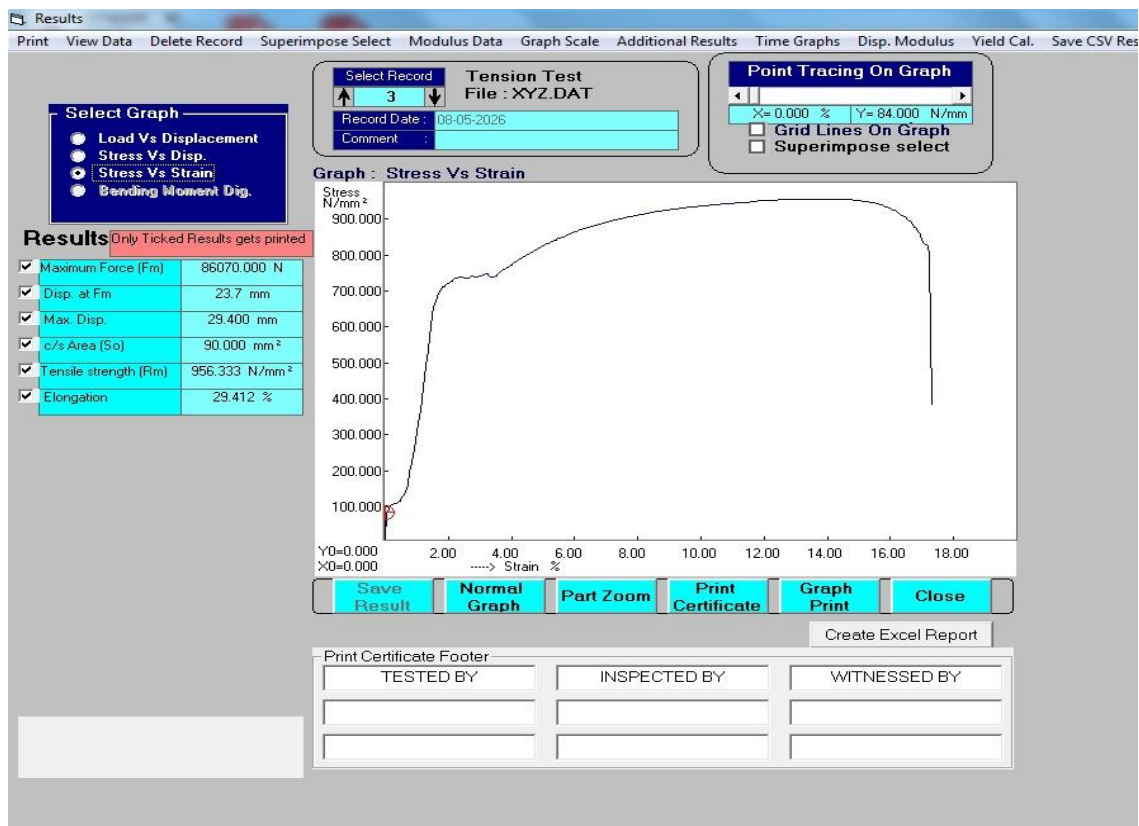


Fig 10: - Stress Vs Strain Graph

XI. MACHINE LEARNING: -

Mild Steel Tension Test Results | Cross-sectional Area: 90 mm² | Date: 08-05-2026

Sp.	Welding Parameters			Gauge Length (mm)		Mechanical Properties			
No.	Current (A)	Voltage (V)	Gas Flow (L/min)	Before	After	Max Force (N)	Tensile Strength (N/mm ²)	Elongation (%)	Max Disp. (mm)
1	60	73	6	175	205	83,190	924.333	13.889	16.9
2	60	83	8	170	220	84,561	939.619	21.306	23.5
3	60	94	10	175	227	86,070	956.333	29.412	29.4
4	70	97	10	172	220	84,060	934.000	29.714	27.6
5	70	90	8	173	203	83,280	925.333	27.907	28.3
6	70	83	6	172	225	84,030	933.667	23.121	23.0
7	80	98	6	—	—	82,200	913.333	16.667	11.9
8	80	97	8	175	225	84,510	939.000	28.571	29.5
9	80	83	10	176	230	82,770	919.667	30.682	34.7
Statistical Summary (all 9 specimens)									
Average						83,818	931.310	23.475	24.5
Maximum						86,070	956.333	30.682	34.7
Minimum						82,200	913.333	13.889	11.9

Table 1- Results from UTM tests

The Machine Learning model is developed using these results obtained after performing tests on Ultimate Testing Machine.

Machine Learning Model Used:

A Random Forest Regression model was used to predict the mechanical properties of TIG welded mild steel specimens based on welding process parameters. The model predicts tensile strength, elongation, stress, and maximum force using welding current, arc voltage, and gas flow rate as input features.

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each decision tree is trained using randomly selected subsets of the dataset, and the final prediction is obtained by averaging the outputs of all trees.

In this work, separate Random Forest Regressor models were trained for each mechanical property. The model was implemented using the Scikit-learn machine learning library in Python.

Why This Model Was Selected?

The Random Forest Regression model was selected because it can capture complex non-linear relationships between welding parameters and mechanical properties without requiring explicit mathematical equations.

Unlike simple linear regression models, Random Forest can automatically learn interactions between current, voltage, and gas flow rate. The model is also resistant to noise and provides stable predictions even when experimental data contain variations.

The main advantages of Random Forest for this application are:

1. Ability to model non-linear welding behaviour

Welding properties are influenced by complex thermal and metallurgical interactions. Random Forest can capture these non-linear effects effectively.

2. Reduced overfitting

By averaging multiple decision trees, the model improves generalisation and reduces prediction variance.

3. No requirement for explicit equations

The algorithm learns patterns directly from experimental data.

4. Good prediction capability

Random Forest performs well for engineering property prediction problems where relationships between parameters are highly coupled.

5. Capability to predict properties without repeated destructive testing

Once trained, the model can estimate mechanical properties directly from welding parameters, reducing the need for repeated UTM testing.

Explanation of the Code:

The developed Python program first stores the experimental welding dataset containing current, voltage, gas flow rate, tensile strength, elongation, and maximum force values obtained from tensile testing. The input parameters are combined into a feature matrix and used to train separate Random Forest Regression models for predicting each mechanical property. The Scikit-learn RandomForestRegressor algorithm was used with 200 decision trees to improve prediction stability and accuracy.

After training, the program accepts new welding parameter values from the user. The trained machine learning models then predict the corresponding tensile strength, elongation, stress, and maximum force values without requiring additional destructive testing.

The final predicted mechanical properties are displayed automatically for the entered welding conditions.

XII RESULTS: -

TIG WELDING PROPERTY PREDICTOR		
Current	: 60 – 80	A
Voltage	: 73 – 98	V
Gas Flow	: 6 – 10	L/min
Enter Current (A) : 63		
Enter Voltage (V) : 78		
Enter Gas Flow (L/min) : 8		
PREDICTED MECHANICAL PROPERTIES		
Tensile Strength	: 934.371	N/mm ²
Elongation	: 21.921	%
Stress	: 934.294	N/mm ²
Max Force	: 84086.5	N

Fig 11- Result for given input parameters

1. Mild steel plates of size 30 mm × 150 mm × 3 mm were successfully cut and prepared.
2. A total of 11 specimens were welded using the TIG welding process.
3. Welding parameters such as current (60–80 A), voltage (73–98 V), and gas flow rate (6–10 L/min) were varied.
4. All welded specimens were tested using a Universal Testing Machine (UTM).
5. Load vs Displacement and Stress vs Strain graphs were obtained for each specimen.
6. Mechanical properties such as:
 - Maximum Force (N)
 - Tensile Strength (MPa)

- Percentage Elongation (%)
 - Displacement (mm) were extracted from the test results
7. The experimental data were arranged into a dataset for machine learning.
 8. Machine learning and deep learning models were developed using the dataset.
 9. . The models successfully predicted the mechanical properties of TIG-welded joints.
 10. The project demonstrated that machine learning can reduce the need for repeated tensile testing and help optimize welding parameters.

XV. CONCLUSIONS

1. Mild steel plates of dimensions 30 mm × 150 mm × 3 mm were successfully prepared and welded using the TIG welding process.
2. Eleven welded specimens were fabricated using different combinations of welding current, voltage, and shielding gas flow rate.
3. Tensile testing was carried out using a Universal Testing Machine (UTM).
4. Load–Displacement and Stress–Strain curves were obtained for all specimens.
5. Mechanical properties such as maximum force, tensile strength, and percentage elongation were extracted.
6. The experimental data were used to create a dataset for machine learning and deep learning models.
7. The developed models successfully predicted the mechanical properties of TIG-welded joints.
8. The project showed that machine learning can reduce the need for repeated destructive testing and help optimize welding parameters
9. The methodology is cost-effective and can be extended to aerospace and other engineering applications.

XIV. SUGGESTIONS FOR FURTHER WORKS: -

1. Increase the number of welded specimens to create a larger dataset and improve the accuracy of the machine learning models.
2. Include additional welding parameters such as welding speed, electrode angle, and filler material type.
3. Perform other mechanical tests such as hardness, impact, and bend tests to obtain more comprehensive material properties.
4. Apply the same methodology to advanced materials such as stainless steel, aluminum alloys, and aerospace-grade steels.
5. Develop more advanced deep learning models and optimization algorithms to improve prediction performance.
6. Conduct microstructural analysis using optical microscopy or SEM to correlate weld structure with mechanical properties.
7. Develop a user-friendly software or web application for predicting material properties from welding parameters.
8. Integrate the model with real-time welding systems for online monitoring and quality prediction.
9. Compare the performance of different machine learning algorithms on larger datasets.
10. Extend the study to other welding processes such as MIG, laser welding, and friction stir welding.

XV APPENDIX

CODE:

```
import numpy as np

from sklearn.ensemble import RandomForestRegressor

# — Training Data (9 TIG welded specimens) —
current = [60, 60, 60, 70, 70, 70, 80, 80, 80]
voltage = [73, 83, 94, 97, 90, 83, 98, 97, 83]
gas_flow = [ 6, 8, 10, 10, 8, 6, 6, 8, 10]

tensile = [924.333, 939.619, 956.333, 934.000, 925.333, 933.667,
913.333, 939.000, 919.667]
elongation= [13.889, 21.306, 29.412, 29.714, 27.907, 23.121, 16.667,
28.571, 30.682]
force = [83190, 84561, 86070, 84060, 83280, 84030, 82200, 84510,
82770]
stress = [f / 90 for f in force] # Stress = Force / Cross-section (90
mm2)

X = list(zip(current, voltage, gas_flow))

# — Train one model for each output —
model_tensile = RandomForestRegressor (n_estimators=200,
random_state=42). fit (X, tensile)
```

```

model_elongation = RandomForestRegressor(n_estimators=200,
random_state=42).fit(X, elongation)
model_force      = RandomForestRegressor(n_estimators=200,
random_state=42).fit(X, force)
model_stress     = RandomForestRegressor(n_estimators=200,
random_state=42).fit(X, stress)

```

```

# — Take Input —

```

```

print("=" * 50)
print("  TIG WELDING PROPERTY PREDICTOR")
print("=" * 50)
print(" Current   : 60 – 80  A")
print(" Voltage   : 73 – 98  V")
print(" Gas Flow   : 6 – 10  L/min")
print("=" * 50)

```

```

cur = float(input("\n Enter Current (A)   : "))
vol = float(input(" Enter Voltage (V)     : "))
gas = float(input(" Enter Gas Flow (L/min) : "))

```

```

# — Predict —

```

```

inp = [[cur, vol, gas]]

t = model_tensile.predict(inp)[0]
e = model_elongation.predict(inp)[0]

```

```

f = model_force.predict(inp)[0]
s = model_stress.predict(inp)[0]

# — Display Output —
print("\n" + "=" * 50)
print("  PREDICTED MECHANICAL PROPERTIES")
print("=" * 50)
print(f"  Tensile Strength : {t:.3f} N/mm2")
print(f"  Elongation      : {e:.3f} %")
print(f"  Stress          : {s:.3f} N/mm2")
print(f"  Max Force       : {f:.1f} N")
print("=" * 50)

```

XVI COURSE – PO CORRELATION MATRIX

Course: Theme based Project		Course code: U23PW619ME								Sem.: VI			
Course Outcomes: On completion of the course, the student will be able to													
CO1	Identify appropriate field of interest, review the literature and define the problem.												
CO2	Plan the activities for carrying out the research work in teams to solve the identified problem using different resources.												
CO3	work in teams and adapt for the advanced technological changes												
CO4	Conduct the investigations on the chosen problem and prepare the final report.												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
1													
2													
3													
4													
Avg.													
CO	PSO1	PSO2	PSO3										
1													
2													
3													
4													
Avg.													

Program Specific Outcomes (PSOs)	
1	apply principles of Basic sciences and engineering to mechanical systems.
2	model, analyze, design, and realize Mechanical components and processes.
3	be prepared to work professionally and ethically in thermal, design and manufacturing areas of mechanical engineering.

Program Outcomes (POs)	
1	Graduates demonstrate knowledge of basic sciences and mechanical engineering.
2	Graduates demonstrate an ability to identify, formulate and solve engineering problems
3	Graduates demonstrate an ability to design and conduct experiments, analyze and interpret data.
4	Graduates demonstrate an ability to design a system, component or process as per needs and specifications
5	Graduates demonstrate skills to use modern engineering tools, software and equipment to analyze for problem solving.
6	Graduates demonstrate an ability to visualize and work on laboratory and multi disciplinary tasks.
7	Graduate shows the understanding of impact of environment and society of engineering solutions and aim to provide sustainable solutions.
8	Graduates demonstrate knowledge of professional and ethical responsibilities.
10	Graduates are able to communicate effectively in both verbal and written form.
11	Graduates will demonstrate the ability to handle the projects through appropriate project management techniques.
12	Graduates develop confidence for self education and ability for life-long learning

XVII REFERENCES

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